University College London, 2025-2026
Econ 0016 – International Macroeconomics – Franck Portier

Problem set 1 : Global Imbalances, Current Account Sustainability and Intertemporal Theory of the Current Account

Chapter 1, 2 and 3 in Schmitt-Grohé, Uribe and Woodford

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Exercise 1.2 (Balance of Payments Accounting)

Describe how each of the following transactions affects the U.S. balance of payments. (Recall that each transaction gives rise to two entries in the balance of payments accounts.)

1. Jorge Ramírez, a landscape architect residing in Monterrey, Mexico, works for three months in Durham, North Carolina, creating an indoor garden for a newly built museum and receives wages of \$35,000.
2. Jinill Park's mother, a resident of South Korea, pays her son's \$60,000 tuition to Columbia University via a direct deposit.
3. Columbia University buys several park benches from Spain and pays with a \$120,000 check.
4. Floyd Townsend, of Tampa, Florida, buys \$5,000 worth of British Airlines stock from Citibank New York, paying with U.S. dollars.
5. A group of American friends travels to Costa Rica and rents a vacation home for \$2,500. They pay with a U.S. credit card.
6. The United States sends medicine, blankets, tents, and nonperishable food worth \$400 million to victims of an earthquake in a foreign country.

Exercise 1.6 (Balance of Payments)

This question is about the balance of payments of a country named Outland. The currency of Outland is the dollar.

1. Outland starts a given year with holdings of 100 shares of the German car company Volkswagen. These securities are denominated in euros. The rest of the world holds 200 units of dollar-denominated bonds issued by the Outlandian government. At the beginning of the year, the price of each Volkswagen share is €1 and the price of each unit of an Outlandian bond is \$2. The exchange rate is \$1.5 per euro. Compute the net international investment position (NIIP) of Outland at the beginning of the year.

2. During the year, Outland exports toys for \$7 and imports shirts for €9. The dividend payments on the Volkswagen shares were €0.05 per share and the coupon payment on Outlandian bonds was \$0.02 per bond. Residents of Outland received money from relatives living abroad for a total of €3 and the government of Outland gave \$4 to a hospital in Guyana. Calculate the Outlandian trade balance, net investment income, and net unilateral transfers in that year. What was the current account in that year? What is the Outlandian NIIP at the end of the year?

3. Suppose that at the end of the year, Outland holds 110 Volkswagen shares. How many units of Outlandian government bonds are held in the rest of the world? Assume that during the year, all financial transactions were performed at beginning-of-year prices and exchange rates.

4. Let's move to the next year and start with the international asset and liability positions calculated in item 3. Suppose that at the end of the year, the price of a Volkswagen share falls by 20 percent and the dollar appreciates by 10 percent while the current account is 0 for that year. Calculate the end-of-year NIIP of Outland.

Exercise 1.12 (Dark Matter versus Return Differentials)

Suppose net investment income is NII = 200, the international asset position is A = 3000, the international liability position is L = 4000, and the rate of return is 5 percent, $r = 0.05$.

1. Economist John Green, a strong advocate of the dark matter hypothesis, believes that A is not accurately recorded. Calculate the amount of dark matter and the "true" international asset position, which we will denote TA, consistent with Green's view.

2. Financial analyst Nadia Gonzalez does not believe in the dark matter hypothesis. Instead, she believes that A is accurately measured. In her view 5 percent is actually the rate of return on assets $r^A = 0.05$, and the rate of return on the country's international liabilities, r^L, is different. Find the value of r^L consistent with Gonzalez's view.

Exercise 2.3 (Saving, Investment, and the Net International Investment Position)

In a two-period economy, saving in periods 1 and 2 is 5 ($S_1 = S_2 = 5$) and investment in both periods is 10 ($I_1 = I_2 = 10$).

1. What is the current account in periods 1 and 2 (CA_1 and CA_2)?

2. What is the initial net international investment position (B_0)?

3. Assuming that the interest rate is 4 percent ($r = 0.04$), what is the trade balance in periods 1 and 2 (TB_1 and TB_2)?

Notes: (Log Utility, Transitory and Permanent Endowment Shocks)

These notes present the explicit algebra for solving the intertemporal model of the current account when preferences are log additive. Make sure you can replicate all the steps in the derivation on the results.

We consider here a two-period lived household, whose preferences for consumption are described by the log additive lifetime utility function:

$$\log C_1 + \log C_2$$

where C_1 and C_2 denote consumption in periods 1 and 2, respectively. In the log additive case, it happens that analytical expression are quite nice and simple. let's derive them.

1– Let us first check that these preferences give rise to indifference curves that are downward sloping and convex. An indifference curve has equation $\log C_1 + \log C_2 = \overline{U}$ for arbitrary $\overline{U}$, which can be rewritten as

$$C_2 = f(C_1) = e^{\overline{U} - \log C_1}$$

The indifference curve is decreasing convex if $f'(\cdot) < 0$ and $f(\cdot)'' > 0$. Indeed,

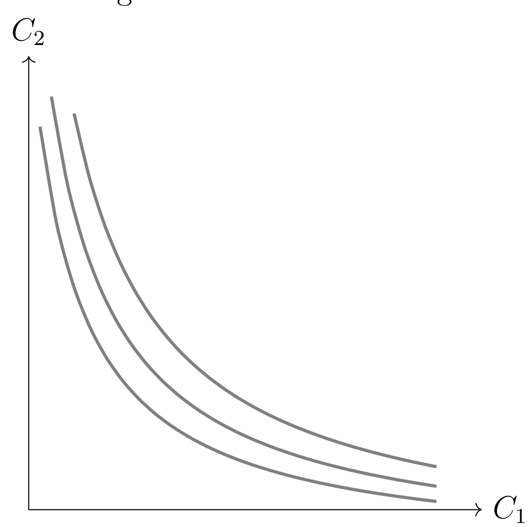
$$f'(C_1) = -e^{\overline{U} - \log C_1} \frac{1}{C_1} < 0$$

and

$$f''(C_1) = e^{\overline{U} - \log C_1} \frac{1}{C_1^2} + e^{\overline{U} - \log C_1} \frac{1}{C_1^3} > 0.$$

Furthermore, one can check that $\lim_{C_1 \to 0} f'(C_1) = -\infty$ and $\lim_{C_1 \to \infty} f'(C_1) = 0$.

Figure: Indifference curves



2– Suppose that the household starts period 1 with financial wealth equal to $(1 + r_0)B_0$, where B_0 is an inherited stock of bonds and r_0 is the interest rate on assets held between periods 0 and 1. In addition, the household receives endowments of goods in the amounts Q_1 and Q_2 in periods 1 and 2, respectively. In period 1, the household can borrow or lend at the interest rate $r_1 > 0$ via a bond denoted B_1. Let's find the optimal levels of consumption in periods 1 and 2 as functions of the household's lifetime wealth, $\overline{Y} = (1 + r_0)B_0 + Q_1 + Q_2/(1 + r_1)$ and the interest rate r_1.

The period-1 budget constraint is

$$C_1 + B_1 - B_0 = r_0 B_0 + Q_1. \tag{1}$$

and the period-2 budget constraint is

$$C_2 + B_2 - B_1 = r_1 B_1 + Q_2. \tag{2}$$

Because the world ends after period 2, no one is going to be around to pay or collect debts. So bond holdings must be nil at the end of period 2, that is,

$$B_2 = 0. \tag{3}$$

We denoted this expression as the *transversality condition*.

Combining (1), (2) and (3) to eliminate B_1 and B_2, we obtain the the intertemporal budget constraint (IBC)

$$C_1 + \frac{C_2}{1+r_1} = \underbrace{(1+r_0)B_0 + Q_1 + \frac{Q_2}{1+r_1}}_{\overline{Y}}. \tag{4}$$

This IBC It says that the present discounted value of the endowment plus the initial financial wealth (the right-hand side) must be equal to the present discounted value of consumption (the left-hand side).

The IBC implies $C_2 = (1+r_1)(\overline{Y} - C_1)$, so that the maximisation problem of the representative agent is:

$$\max_{C_1} \ \log C_1 + \log\Big((1+r_1)(\overline{Y} - C_1)\Big)$$

or

$$\max_{C_1} \ \log C_1 + \log(\overline{Y} - C_1) + \log(1+r_1)$$

First order condition is

$$\frac{1}{C_1} - \frac{1}{\overline{Y} - C_1} = 0$$

which implies

$$C_1 = \frac{1}{2}\overline{Y}$$

and then

$$C_2 = \frac{1}{2}(1+r_1)\overline{Y}$$

We can then compute the trade balance and the current account in period 1.

$$\begin{aligned}
TB_1 &= Q_1 - C_1 \\
&= Q_1 - \frac{1}{2}\overline{Y} \\
&= Q_1 - \frac{1}{2}\Big((1+r_0)B_0 + Q_1 + \frac{Q_2}{1+r_1}\Big) \\
&= \frac{1}{2}\Big(Q_1 - (1+r_0)B_0 - \frac{Q_2}{1+r_1}\Big).
\end{aligned}$$

and

$$
\begin{aligned}
CA_1 &= TB_1 + r_0 B_0 \\
&= \frac{1}{2}\left(Q_1 - (1+r_0)B_0 - \frac{Q_2}{1+r_1}\right) + r_0 B_0 \\
&= \frac{1}{2}\left(Q_1 - (1-r_0)B_0 - \frac{Q_2}{1+r_1}\right).
\end{aligned}
$$

3- Let's now compute the responses of consumption in period 1, ΔC_1, the trade balance in period 1, ΔTB_1, and the current account in period 1, ΔCA_1, to a temporary increase in the endowment, $\Delta Q_1 = \Delta Q > 0$ and $\Delta Q_2 = 0$.

In that case, we have $\Delta \overline{Y} = \Delta Q$, and therefore

$$
\begin{aligned}
\Delta C_1 &= \frac{1}{2}\Delta Q \\
\Delta TB_1 &= \Delta Q - \Delta C_1 \\
&= \frac{1}{2}\Delta Q \\
\Delta CA_1 &= \Delta TB_1 \\
&= \frac{1}{2}\Delta Q
\end{aligned}
$$

4– Let's move a a permanent shock and find the responses of consumption in period 1, ΔC_1, the trade balance in period 1, ΔTB_1, and the current account in period 1, ΔCA_1, to a permanent increase in the endowment, $\Delta Q_1 = \Delta Q_2 = \Delta Q > 0$.

In that case, we have $\Delta \overline{Y} = \left(1 + \frac{1}{1+r_1}\right)\Delta Q > \Delta Q$, and therefore

$$
\begin{aligned}
\Delta C_1 &= \frac{1}{2}\left(1 + \frac{1}{1+r_1}\right)\Delta Q \\
\Delta TB_1 &= \Delta Q - \Delta C_1 \\
&= \frac{1}{2}\left(1 - \frac{1}{1+r_1}\right)\Delta Q \\
\Delta CA_1 &= \Delta TB_1 \\
&= \frac{1}{2}\left(1 - \frac{1}{1+r_1}\right)\Delta Q
\end{aligned}
$$